

# KICKINGEREDER ET AL. (2016)

Radiomics of brain MRI: machine learning-based classification of molecular subtypes in glioblastoma

Radiology

▢ Citations: 1,800+

▢ Impact Factor: 11.105

▢ Category: Neuro-oncology

## ABSTRACT SUMMARY

▢ Key Contribution: Molecular prediction in glioblastoma

• First comprehensive radiomics study in glioblastoma

• Correlation with molecular markers (MGMT, IDH1)

• Prognostic value in brain tumors

• Integration with clinical factors

• Multi-parametric MRI analysis

# METHODOLOGY ANALYSIS

## □ STUDY DESIGN

• Dataset: 119 glioblastoma patients

• Imaging: Multi-parametric MRI (T1, T2, FLAIR, DWI)

• Features: 1,200+ radiomics features extracted

• Molecular: MGMT methylation, IDH1 mutation status

## □ MULTI-PARAMETRIC ANALYSIS

• Validation: Cross-validation and independent testing

• T1-weighted: Anatomical and contrast enhancement

• T2-weighted: Edema and tumor extent

• FLAIR: Peritumoral edema assessment

• DWI: Cellularity and diffusion characteristics

• Feature Integration: Cross-modality analysis

# KEY RESULTS & FINDINGS

## MOLECULAR PREDICTION RESULTS

- MGMT Methylation: AUC = 0.85 (high accuracy)
- IDH1 Mutation: Significant correlation with radiomics
- Feature Selection: LASSO regression for feature selection
- Validation: Cross-validation confirmed results

## PROGNOSTIC VALUE

- Clinical Integration: Combined with clinical factors
- Progression-free survival prediction
- Overall survival correlation
- Risk stratification capabilities
- Treatment response prediction
- Clinical decision support

# CLINICAL IMPACT & INNOVATION

## □ CLINICAL IMPACT

• Non-invasive molecular prediction

• Personalized treatment planning

• Reduced need for invasive biopsies

• Improved prognostic assessment

## □ INNOVATION

• Clinical trial stratification

• First glioblastoma radiomics study

• Multi-parametric MRI integration

• Molecular marker prediction

• Clinical-radiomics integration

• Neuro-oncology applications

# LEGACY & FUTURE DIRECTIONS

## □ □ LEGACY

• Foundation for neuro-oncology radiomics

• Multi-parametric MRI standard

• Molecular prediction methodology

• Clinical integration framework

## □ FUTURE DIRECTIONS

• Glioblastoma treatment planning

• Clinical implementation in glioblastoma

• Extension to other brain tumors

• Real-time molecular prediction

• Treatment response monitoring

• Clinical trial integration